

Fig. 1 | Cerebellar involvement in motor planning and movement initiation.

a, Mouse reporting the location of a pole by directional licking. **b**, The ALM-cerebellar loop. **c**, Top, schematic of a fastigial or a dentate lesion. Bottom, example lesion. **d**, Performance of mice in the delayed response task (session mean $\pm$ s.e.m., $n = 75$ –256 trials). The numbers at the top of the graphs refer to the number assigned to each individual mouse. Mouse 1, 2, left fastigial lesion; mouse 3, 4, right fastigial lesion; mouse 5, left dentate lesion; mouse 6–8, right dentate lesion. Lick-left and lick-right trials are grouped by instructed licking direction relative to the lesioned hemisphere. Blue, contralesional (contra); red, ipsilesional (ipsi). *** $P < 0.001$, one-sided test, pre- versus post lesion, bootstrap (Methods). **e**, Top, lick rate during contra- and ipsilesional licking bouts after fastigial lesion. Lick times are aligned to the first lick ($t = 0$). Bottom, time of the first lick to the contra- and ipsilesional direction after onset of the 'go' cue. Lines represent individual mice ($n = 4$). *** $P < 0.001$, one-sided test, bootstrap. **f**, Top, schematic of fastigial or dentate photoactivation. Laser power, 0.5–1.5 mW. Bottom, example ChR2 expression. **g**, Performance in the delayed response task after fastigial or dentate photoactivation during the delay period. Lick-left and lick-right trials are pooled. Chance is 50% (dashed line). Lines represent individual mice (fastigial, $n = 6$; dentate, $n = 8$); bars represent the mean. *** $P < 0.001$, one-sided test, bootstrap. **h**, Performance in the delayed response task after fastigial photoactivation during specific periods. Lick-left and lick-right trials are grouped by instructed licking direction relative to the manipulated hemisphere. Both hemispheres were tested. See Supplementary Table 1 for manipulated hemispheres. Thick lines represent the mean; thin lines represent individual mice ($n = 6$). Photoact., photoactivation. * $P < 0.05$, *** $P < 0.001$, one-sided test, bootstrap. **i**, Lick rate in control and fastigial photoactivation trials ($n = 6$ mice). **j**, Left, schematic of CN photoinhibition. Laser power, 0.5–4.5 mW. Right, silicon-probe recording in the fastigial nucleus. An example voltage trace and the average firing rate of CN neurons are shown ($n = 26$ neurons, 2 mice). Firing rates were normalized to the baseline (500 ms before photostimulus onset). **k**, Performance in the delayed response task after CN photoinhibition. Experimental details as in **h**. $n = 7$ mice for sample and delay period photoinhibition; $n = 3$ for response period photoinhibition (Supplementary Table 1). **l**, Lick rate in control and CN photoinhibition trials ($n = 3$ mice).

